

An hourglass that reads the past through a cone it shapes itself, tested three ways against its own fixed twin

The design • slides 3-7

The present is a **waist** of ~13 cells, not a tip; the past cone reads the last 30 days at every pentad; the **prediction cone** — the past cone mirrored through the anchor, +1...+6 pentads — is sampled the same way and used only as targets. Each example draws one of **four tasks** (0.35 forecast • 0.15 from the waist alone • 0.25 nowcast • 0.25 fill-in), under E-069b's rules: knowable targets only, no copies, every family against its own predict-the-mean bar

The cone the model shapes • slides 8-12

A fixed 24-point sunflower, phase-rotated by the golden angle per lag, warped by **eight dimensionless numbers per 1° cell and channel group**: a drift **d** toward the source (capped at cone v2's design speed) and two ellipses stored as **matrix logarithms** — no kilometres, no angle to wrap — with $\Sigma(\ell) = \Sigma_0 + \ell^2 \Sigma_v$. **One direction per flow, one scale and one memory per channel**: s_c and τ_c inflate or shrink the shared ellipse and set how far back the channel reads, applied as a Gaussian **aperture** over the group's shared per-location tokens — zero extra tokens. Learned **gate-first** (a soft aperture over fixed dots, so the loader never depends on live weights), then **hardened**

The guards • slides 13, 15

Coarse-10°-plus-1°-residual prior; log-space soft floors; mirror tie; stop-gradient on target positions; evaluation geometry frozen for every arm; the far ring kept and never gated; the aperture never touches the loss weights — so the cone cannot learn to ask easy questions

The experiment • slide 17

Pre-registered. **A0** fixed hourglass (control, and the eval geometry for all) • **A1** learned from A0's init • **A2** learned, surface drift primed at $-u_{clim} \cdot \Delta t$. Same 7 M codec, 20 k steps, frozen protocol, **3 seeds each**, on the same tensor (**family 7.2**); read-outs R1-R5. **Verdict**: A1 beats A0 on R1 at every lead ≤ 3 , paired at all seeds, R5 clean → adopted; A2 > A1 ≈ A0 → geometry from climatology; neither → keep A0

RESULTS — figure to be inserted

R1 • future-cone loss per lead, A0 fixed / A1 learned / A2 primed, 3 seeds • R3 • learned drift vs upstream • R4 • the western-boundary depth check

Reserved for the measured result; stays empty until a run has produced it.